Study Guide: Software and Operating Systems

Name: _____

Period: _____

Intro to Computer Science | Quiz: Thursday, October 22

> **What's on the quiz**
>
> Everything on this study guide, and nothing else. 4 short real-life stories, 50 points total, plus a +10 bonus. Read the story, find the clue, pick the best answer. No notes. The practice questions at the end look just like the quiz.

1. Hardware vs. Software

Hardware = the parts you can touch (RAM, mouse, monitor). **Software** = the instructions that make hardware DO stuff (Chrome, Windows, Minecraft).

Quick test: Can you touch it? Then it's hardware. Hardware is what the computer IS. Software is what it DOES.

2. The Two Teams of Software

	System Software	Application Software (apps)
What it does	Runs and manages the COMPUTER, in the background	Helps YOU do a specific task
Ask yourself	Does it serve the computer?	Does it serve me?
Examples	Operating systems, drivers, utilities	Chrome, YouTube, Minecraft, Google Docs, Spotify, Calculator
School match	The principal and office staff	Your classes: art, PE, CS

The System Software Crew

	What it does	Examples	School match
Operating System (OS)	The computer's manager. Connects you, your apps, and the hardware.	Windows, macOS, Android, iOS, Linux	The principal
Device Driver	The translator. Helps the OS talk to one piece of hardware.	Printer driver, keyboard driver, graphics card driver	An interpreter
Utility	The cleanup and security team. Keeps the computer healthy.	Antivirus, Disk Cleanup, backup tools	Custodians and security

Remember: New printer or device won't work? Check the DRIVER. Apps need an operating system to run. No OS = nothing works.

3. The Operating System's 4 Jobs

You ask. The OS coordinates. The hardware does it.

Job	What the OS does	School match	Real-life example
1. Manage Hardware	Decides who gets to use the CPU, RAM, storage, printer, and screen. Shares it fairly.	Assigning rooms and equipment	Sending your essay to the printer
2. User Interface	Gives you a way to talk to the computer. GUI = clicking icons and menus. CLI = typing commands.	The front desk	Clicking the Chrome icon
3. Manage Files	Creates, saves, names, moves, finds, and protects your files and folders.	The records office	Saving your essay in a folder, renaming a photo
4. Manage Programs	Loads apps into RAM and shares CPU time so many apps run at once.	The bell schedule	Spotify playing while you play a game

What happens when you open an app? (1) You click it: Interface. (2) The OS finds the program: Files. (3) The OS loads it into RAM: Programs. (4) The CPU runs it: Hardware. (5) You see it on screen: Interface. Four jobs, one click, less than one second!

4. Common Operating Systems

Every OS does the same 4 jobs, just on different devices.

Operating System	Device
Windows	Most PCs and laptops (Dell, HP, Lenovo)
macOS	Mac computers (MacBook, iMac)
iOS	iPhones
Android	Samsung phones and many others
Linux	Servers (most websites run on Linux)

5. If You See This... Think This

If the story says...	Think...
"Can you touch it?"	Yes = hardware. No = software.
Helps YOU do a task (games, music, writing, browsing)	Application software

Works in the background to run the computer	System software
New device plugged in, but it doesn't work	Driver is missing
Antivirus, cleaning junk files, backups	Utility
Clicking icons, tapping, typing commands	OS Job 2: User Interface
Saving, naming, moving, or finding a file	OS Job 3: Manage Files
Many apps open at once, or one app freezes the others	OS Job 4: Manage Programs
Who gets the CPU, RAM, printer, or screen	OS Job 1: Manage Hardware

Practice Questions

Try these on your own, then check your answers with Mr. Rana. They look just like the quiz.

1. Write H for HARDWARE or S for SOFTWARE. (Sort)

___ Instagram ___ RAM ___ Windows 11 ___ Mouse ___ Printer driver

2. Leo plugs a brand-new webcam into his laptop. The laptop says "Device not recognized." What is most likely missing? (Multiple choice)

A. More RAM
B. The webcam's driver
C. A new app
D. A bigger screen

Answer: _____

3. Check ALL the items that are APPLICATION software. (Check all)

☐ Minecraft ☐ Google Docs ☐ YouTube
☐ macOS ☐ Antivirus ☐ Printer driver

4. Kayla is using her laptop. Which OS job is it? Write 1, 2, 3, or 4. (Match)

1 Manage Hardware	2 User Interface	3 Manage Files	4 Manage Programs

____ She taps the Spotify icon

____ She renames a photo and moves it into a folder

____ Spotify keeps playing while she plays Roblox

____ Her essay is sent to the printer

5. Name that OS. Write the operating system next to each device. (Match)

An iPhone: _____ A Samsung phone: _____

A MacBook: _____ A Dell laptop: _____

6. Your friend says, "I don't need an operating system. I only use apps." Is your friend right? Explain why or why not. (Explain)

_____

_____

7. You tap the YouTube app. Number these steps 1 to 5 in the order they happen. (Order)

____ The OS loads YouTube into RAM

____ You tap the YouTube icon

____ The CPU runs YouTube's instructions

____ The OS finds the YouTube program in storage

____ You see YouTube on your screen